8. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60W bulb for use in India is R, the resistance of a 60W bulb for use in USA will be

(a) R (b) $2R$

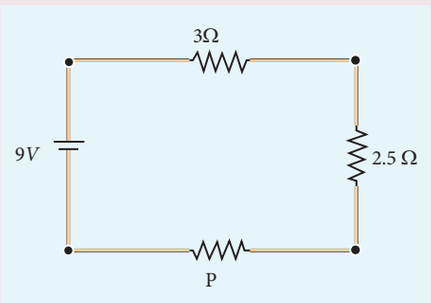
(c) $\frac{R}{4}$ (d) $\frac{R}{2}$

9. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1kW are connected. The voltage of electric mains is 220 V. The maximum capacity of the main fuse of the building will be (IIT-JEE 2014)

(a) 14 A (b) 8 A

(c) 10 A (d) 12 A

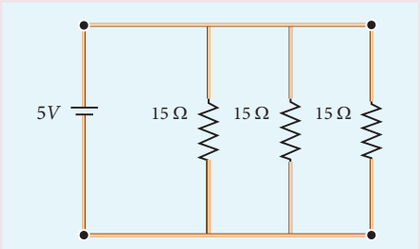
10. There is a current of 1.0 A in the circuit shown below. What is the resistance of P ?



a) 1.5 Ω b) 2.5 Ω

c) 3.5 Ω d) 4.5 Ω

11. What is the current drawn out from the battery?



a) 1A b) 2A

c) 3A d) 4A

12. The temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1 Ω. The resistance of the wire will be 2 Ω at

a) 800 °C b) 700 °C

c) 850 °C d) 820 °C

13. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10 Ω is

a) 0.2 Ω b) 0.5 Ω

c) 0.8 Ω d) 1.0 Ω

14. A piece of copper and another of germanium are cooled from room temperature to 80 K. The resistance of

a) each of them increases

b) each of them decreases

c) copper increases and germanium decreases

d) copper decreases and germanium increases

15. In Joule's heating law, when R and t are constant, if the H is taken along the y axis and I^2 along the x axis, the graph is

a) straight line b) parabola

c) circle d) ellipse

Answers

1) a	**2)** a	**3)** c	**4)** b	**5)** a
6) c	**7)** d	**8)** c	**9)** d	**10)** c
11) a	**12)** d	**13)** b	**14)** d	**15)** a

II Short Answer Questions

1. Why current is a scalar?
2. Define current density.
3. Distinguish between drift velocity and mobility.

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